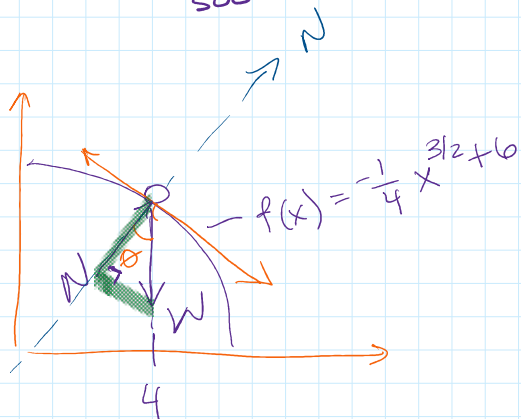
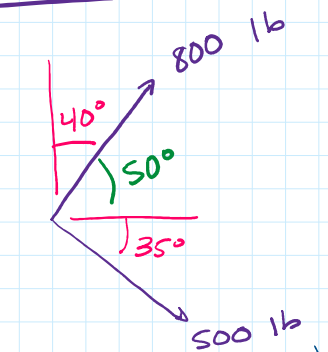


2-10

$$\Sigma F_x = 800 \sin(40^\circ) + 500 \cos(35^\circ)$$

$$\Sigma F_y = 800 \cos(40^\circ) - 500 \sin(35^\circ)$$



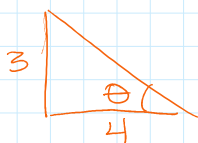
$$\frac{df}{dx} = -\frac{3}{8} x^{1/2}$$

$$\textcircled{a} \ x = 4$$

$$-\frac{3}{8} (4)^{1/2}$$

$$-\frac{3}{8} (2) = \frac{-3}{4}$$

m



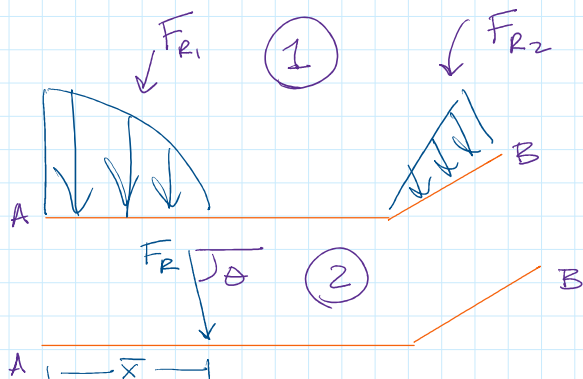
$$\tan \theta = \frac{3}{4} \quad \theta = \tan^{-1}\left(\frac{3}{4}\right) = 36.9^\circ$$

$$\Sigma F_N = N - W \cos \theta = 0$$

$$N = W \cos \theta$$

$$N = (20 \text{ lb}) \cos(36.9^\circ)$$

$$\textcircled{N = 16 \text{ lb}}$$

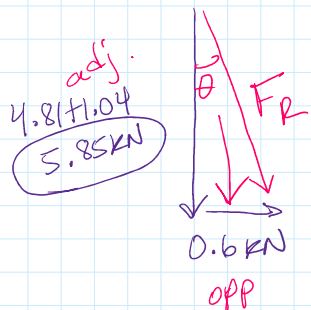
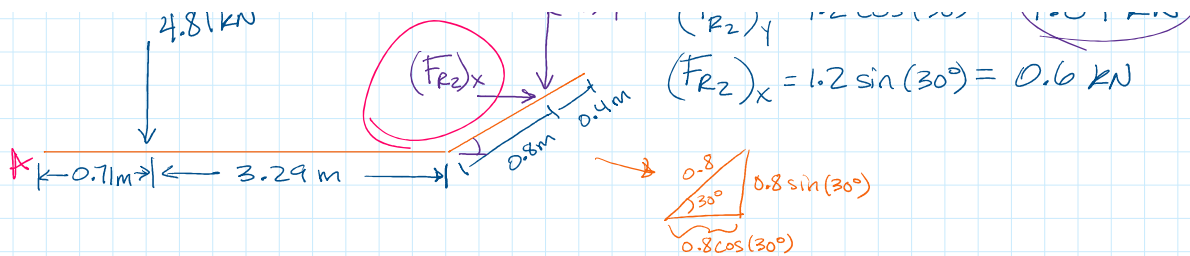


$$4.81 \text{ kN}$$



$$(F_{R2})_y = 1.2 \cos(30^\circ) = \textcircled{1.04 \text{ kN}}$$

$$(F_{R2})_x = 1.2 \sin(30^\circ) = 0.6 \text{ kN}$$

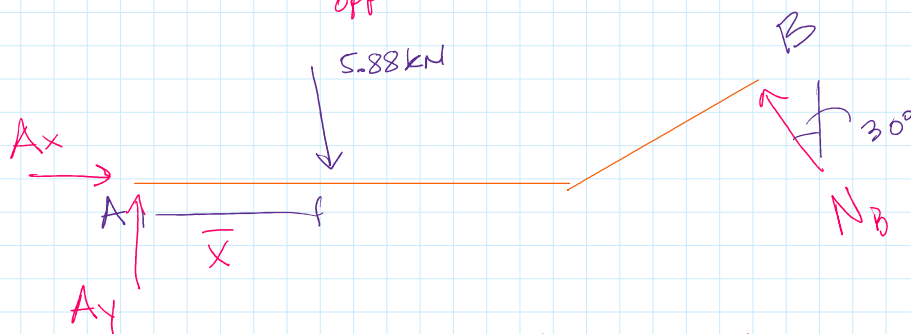


$$F_R = \sqrt{5.85^2 + 0.6^2}$$

$$F_R = 5.88 \text{ kN}$$

$$\theta = \tan^{-1}\left(\frac{0.6}{5.85}\right)$$

$$\theta = 5.9^\circ$$



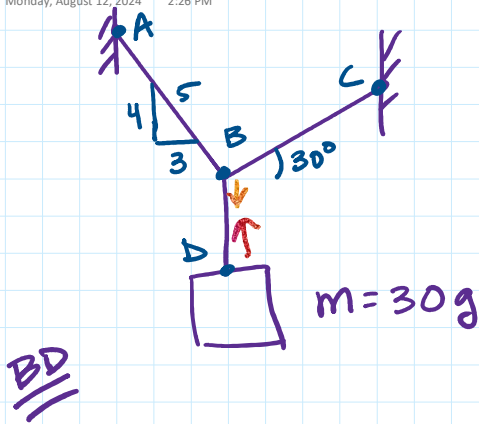
$$(N_B)_x = N_B \sin(30^\circ)$$

$$(N_B)_y = N_B \cos(30^\circ)$$

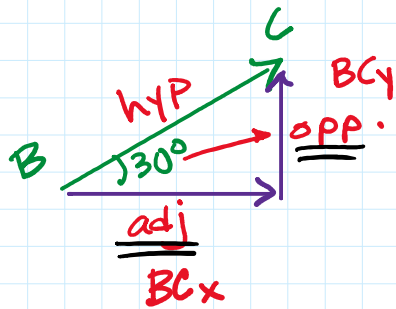
$$\bar{X}(5.85) = (0.71)(4.81) + (4 + 0.8 \cos(30^\circ))(1.04) + (0.8 \sin(30^\circ))(0.6)$$

$$\bar{X} = \frac{3.4151 + 4.88 + 0.24}{5.85}$$

$$\bar{X} = 1.46 \text{ m}$$



* Equilibrium
 $a = 0$
* $\sum \vec{F} = 0$
↑
sigma (sum)



$\vec{BC}$

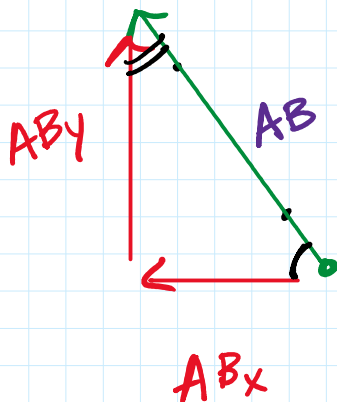
Soh Ca h To a

$(BC) \sin \theta = \frac{BC_y}{BC} (BC)$

$BC_y = BC \cdot \sin \theta$

↓ longest side ↓ < 1

$BC_y = BC \cdot \sin \theta$
 $BC_x = BC \cdot \cos \theta$



* The Δ 's are similar
* Corresponding sides are proportional
* Scale one triangle to get the other one.

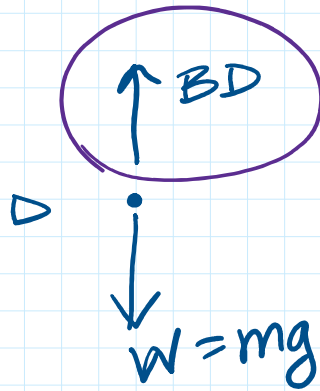
$AB_x = \frac{3}{5} AB = 0.6 AB$

$AB_y = \frac{4}{5} AB = 0.8 AB$



3

FBD



$$\sum F_y = 0$$

$$BD - W = 0$$

$$BD = W$$

$$BD = \underline{mg}$$

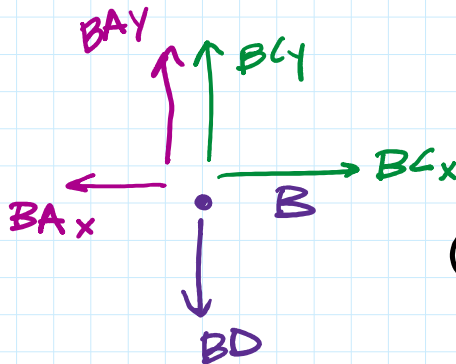
$$BD = (0.03)(9.81)$$

$$BD = 0.29 \text{ N}$$

$$1 \text{ Newton} = 1 \text{ kg} \cdot \text{m/s}^2$$

$$\frac{30 \text{ g}}{1} \left(\frac{1 \text{ kg}}{1000 \text{ g}} \right) = 0.03 \text{ kg}$$

FBD



$$\sum F_x = 0$$

$$BC_x - BA_x = 0$$

$$BC_x = BA_x$$

①

$$BC \cos \theta = 0.6 BA$$

$$\sum F_y = 0$$

$$BA_y + BC_y - BD = 0$$

②

$$0.8 BA + BC \cdot \sin \theta = BD$$

1a

$$BA = \frac{BC \cos(30^\circ)}{0.6} = 1.44 BC$$

$\Rightarrow$

$$BA = 1.44 BC$$

2a

$$0.8 (1.44 BC) + BC \cdot \sin(30^\circ) = 0.29$$

(2a)

$$0.8(1.44 BC) + BC \cdot \sin(30^\circ) = 0.29$$

$$1.15 BC + 0.5 BC = 0.29$$

$$1.65 BC = 0.29$$

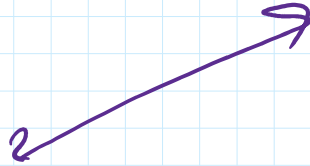
$$BC = 0.175 \text{ N}$$

$$BA = 1.44(0.175 \text{ N})$$

$$BA = 0.25 \text{ N}$$

Brendon

$$BA = 0.26 \text{ N}$$



Right answer

$$\frac{\Delta}{\text{Right}} = \frac{0.01}{0.25} = 0.04 \times 100\% = 4\%$$

3. Equilibrium of a Particle

Tuesday, August 20, 2024 8:42 AM

* Equilibrium: @ rest or constant velocity
 $a=0$

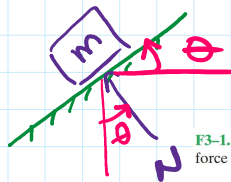
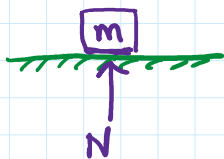
* $\sum \vec{F} = 0$

* Free-Body Diagram (FBD)
 of a point (particle)

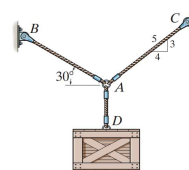
* Spring Force: $F = k \cdot \underline{s} = k(l - l_0)$
 $\downarrow$ stiffness constant $\left[\frac{N}{m}\right]$
 $\uparrow$ unstrained length

* cables in tension

* Normal Force



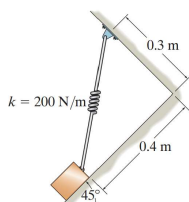
F3-1. The crate has a weight of 550 lb. Determine the force in each supporting cable.



Prob. F3-1

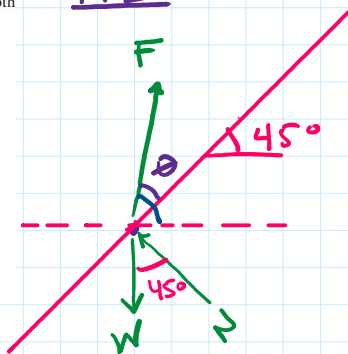
* 2D Space $\sum F_x = 0$, $\sum F_y = 0$
 $\sum F_t = 0$, $\sum F_n = 0$

F3-4. The block has a mass of 5 kg and rests on the smooth plane. Determine the unstretched length of the spring.



Prob. F3-4

FBD



$$\sum F_x = 0$$

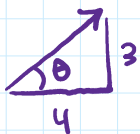
$$\sum F_y = 0$$

$$N_x = N \sin(45^\circ)$$

$$N_y = N \cos(45^\circ)$$

$$F_x = F \cos(\theta + 45^\circ)$$

$$F_y = F \sin(\theta + 45^\circ)$$



$$\tan(\theta) = \frac{3}{4}$$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right) = 36.9^\circ \approx 37^\circ$$

$$\sum F_x = 0 \quad F_x - N_x = 0 \rightarrow F_x = N_x$$

$$\textcircled{1} \quad F \cos(82^\circ) = N \sin(45^\circ) \rightarrow \underline{1a} \quad N = \frac{F \cos(82^\circ)}{\sin(45^\circ)}$$

$$\sum F_y = 0 \quad F_y + N_y - W = 0$$

$$\sum F_x = 0 \quad F_y + N_y - W = 0$$

$$(2) \quad F \sin(82^\circ) + N \cos(45^\circ) = mg$$

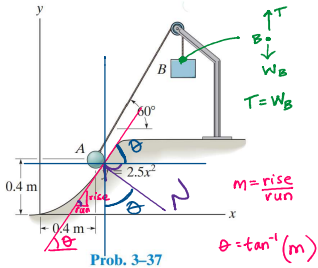
$$F \sin(82^\circ) + \frac{F \cos(82^\circ)}{\sin(45^\circ)} \cos(45^\circ) = mg$$

$$F (\sin(82^\circ) + \cos(82^\circ)) = mg$$

$$F = \frac{mg}{\sin(82^\circ) + \cos(82^\circ)} = \frac{(5 \text{ kg})(9.81 \text{ m/s}^2)}{\sin(82^\circ) + \cos(82^\circ)} = 43.41 \text{ N}$$

$$l_0 = l - \frac{F}{k} = 0.5 - \frac{43.41}{200} = 0.28 \text{ m}$$

3-37. A 4-kg sphere rests on the smooth parabolic surface. Determine the normal force it exerts on the surface and the mass m_B of block B needed to hold it in the equilibrium position shown.

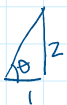


$$y = 2.5x^2$$

* to get slope of tangent line

$$\left. \frac{dy}{dx} \right|_{x=0.4} = 5x = 5(0.4) = 2$$

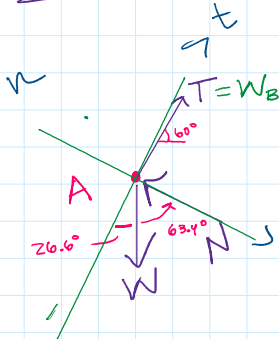
$$m = 2 = \frac{\text{rise}}{\text{run}}$$



$$\theta = \tan^{-1}\left(\frac{2}{1}\right)$$

$$\theta = 63.4^\circ$$

FBD



Equilibrium $\sum \vec{F} = 0$

$$\sum F_x = 0, \quad \sum F_y = 0$$

$$T_x = T \cos(60^\circ) = \frac{1}{2}T = 0.5T$$

$$T_y = T \sin(60^\circ) = \frac{\sqrt{3}}{2}T = 0.866T$$

$$N_x = -N \sin(63.4^\circ) = -0.89N$$

$$N_y = N \cos(63.4^\circ) = 0.44N$$

$$W_x = 0$$

$$W_y = W_A$$

$$\sum F_x = 0 \quad T_x - N_x = 0 \quad (1)$$

$$0.5T - 0.89N = 0$$

$$\sum F_y = 0 \quad T_y + N_y - W = 0 \quad (2)$$

$$0.866T + 0.44N = W_A \rightarrow m_A g$$

$$W_A = (4 \text{ kg})(9.81 \text{ m/s}^2) = 39.2 \text{ N}$$

$$(1a) \quad 0.5T - 0.89N = 0$$

$$0.5T = 0.89N$$

$$T = 1.78 \text{ N}$$

$$(2a)$$

$$0.866(1.78 \text{ N}) + 0.44N = 39.2$$

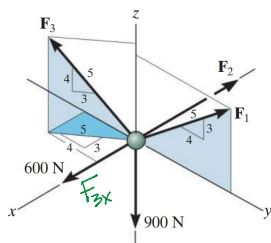
$$1.98 \text{ N} = 39.2$$

$$N = 19.78 \text{ N}$$

$$T = 1.78(19.78) = 35.21 \text{ N}$$

$$m_B = \frac{W_B}{g} = \frac{T}{g} = \frac{35.21 \text{ N}}{9.81 \text{ m/s}^2} = 3.58 \text{ kg}$$

F3-7. Determine the magnitude of forces F_1 , F_2 , F_3 , so that the particle is held in equilibrium.



Prob. F3-7

x, y, z components
of F_1, F_2, F_3

$$F_{1x} = 0 \quad F_{1y} = \frac{4}{5}F_1 \quad F_{1z} = \frac{3}{5}F_1$$

$$F_{2x} = -F_2 \quad F_{2y} = 0 \quad F_{2z} = 0$$

$$F_{3x} = \frac{4}{25}F_3 \quad F_{3y} = -\frac{12}{25}F_3 \quad F_{3z} = \frac{3}{5}F_3$$

$$\sum \vec{F} = 0 \quad \text{Equilibrium}$$

$$\sum F_x = 0 \quad \sum F_y = 0 \quad \sum F_z = 0$$

$$\sum F_x = 0$$

$$F_{1x} + F_{2x} + F_{3x} + 600 = 0$$

$$0 - F_2 + \frac{4}{25}F_3 = -600 \quad (1)$$

$$\sum F_y = 0$$

$$4/5 F_1 + 0 - 12/25 F_3 = 0 \quad (2)$$

$$\begin{bmatrix} 0 & -1 & 4/25 \\ 4/5 & 0 & -12/25 \end{bmatrix} \cdot \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} = \begin{bmatrix} -600 \\ 0 \end{bmatrix}$$

$$\sum F_y = 0$$

$$4/5 F_1 + 0 - \frac{12}{25} F_3 = 0 \quad (2)$$

$$\sum F_z = 0$$

$$3/5 F_1 + 0 + 4/5 F_3 = 900 \quad (3)$$

$$F_{1z} + F_{2z} + F_{3z} - 900 = 0$$

$$\begin{bmatrix} 0 & 1 & 12/25 \\ 4/5 & 0 & -12/25 \\ 3/5 & 0 & 4/5 \end{bmatrix} \cdot \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} = \begin{bmatrix} -600 \\ 0 \\ 900 \end{bmatrix}$$

$$x = A^{-1} b$$

$$(2a) \quad 4/5 F_1 - \frac{12}{25} F_3 = 0$$

$$F_1 = \left(\frac{3}{4}\right) \left(\frac{12}{25} F_3\right)$$

$$F_1 = \frac{3}{5} F_3$$

→ substitute into (3)

Roy

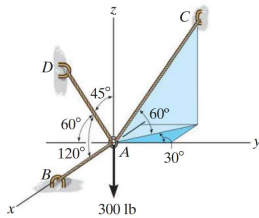
Oscar

Gabriel

Hannah

Ivan

F3-10. Determine the tension developed in cables AB, AC, and AD.



Prob. F3-10

$$AD_x = AD \cos(120^\circ) = -0.5 AD$$

$$AD_y = -AD \cos(60^\circ) = -0.5 AD$$

$$AD_z = AD \cos(45^\circ) = 0.707 AD$$

$$AC_x = AC \cos(60^\circ) \sin(30^\circ) = -0.25 AC$$

$$AC_y = AC \cos(60^\circ) \cos(30^\circ) = 0.433 AC$$

$$AC_z = AC \sin(60^\circ) = 0.866 AC$$

$$AB_x = AB$$

$$AB_y = 0$$

$$AB_z = 0$$

$$\sum F_x = 0$$

$$(1) \quad -0.5 AD - 0.25 AC + AB = 0$$

$$\sum F_y = 0$$

$$(2) \quad -0.5 AD + 0.433 AC + 0 = 0 \rightarrow$$

$$AD = 0.866 AC$$

Substitute

$$\sum F_z = 0$$

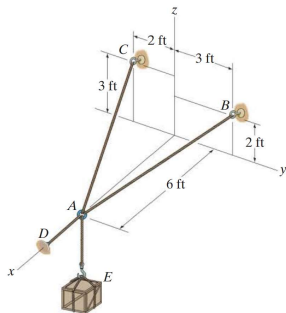
$$(3) \quad 0.707 AD + 0.866 AC + 0 = 300$$

$$AB = 139 \text{ lb}$$

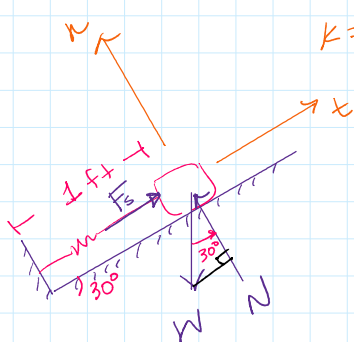
$$AC = 203 \text{ lb}$$

$$AD = 176 \text{ lb}$$

F3-11. The 150-lb crate is supported by cables AB, AC, and AD. Determine the tension in these wires.



Prob. F3-11



$k = ?$

$$W = 50 \text{ lb}$$

$$W = mg - [N] \text{ or } [lb]$$

$$l = 0.5 \text{ ft}$$

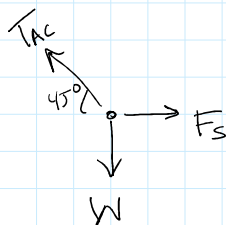
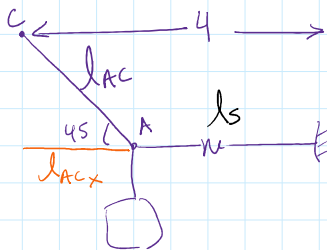
$$l_0 = 1 \text{ ft}$$

$$F_s = k(l - l_0)$$

$$F_s = W \sin(30^\circ)$$

$$k(0.5) = W(0.5)$$

$$k = 50 \text{ lb/ft}$$



l_{AC}

$$T_{AC} \cos(45^\circ) = F_s$$

$$T_{AC} \sin(45^\circ) = W$$

$$W = mg = 20 \text{ kg} (9.81 \text{ m/s}^2)$$

$$F_s = k(l - l_0)$$

$$F_s = W \frac{\cos(45^\circ)}{\sin(45^\circ)}$$

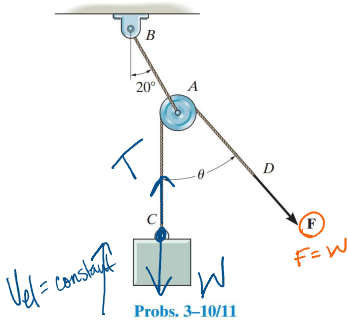
$$l_{AC} \cos(45^\circ) + l_s = 4$$

3-10. The block has a weight of 20 lb and is being hoisted at uniform velocity. Determine the angle θ for equilibrium and the force in cord AB.

$a = 0$ since velocity = const.

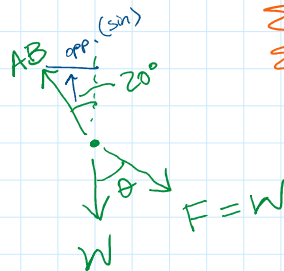
3-10. The block has a weight of 20 lb and is being hoisted at uniform velocity. Determine the angle θ for equilibrium and the force in cord AB.

3-11. Determine the maximum weight W of the block that can be suspended in the position shown if cords AB and CAD can each support a maximum tension of 80 lb. Also, what is the angle θ for equilibrium?



$$a = 0 \text{ since velocity} = \text{const.}$$

FBD @ A



$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$AB_x = -AB \sin(20^\circ)$$

$$AB_y = AB \cos(20^\circ)$$

$$F_x = F \sin \theta$$

$$F_y = F \cos \theta$$

$$\sum F_x = 0$$

$$AB_x + F_x = 0$$

$$-AB \sin(20^\circ) + F \sin \theta = 0$$

$$\textcircled{1} \quad \sin \theta = \frac{AB \sin(20^\circ)}{F}$$

$$AB = \frac{F \sin \theta}{\sin(20^\circ)}$$

$$\sum F_y = 0$$

$$AB_y - F_y - W = 0$$

$$AB \cos(20^\circ) - F \cos \theta - W = 0$$

$$F \cos \theta = AB \cos(20^\circ) - W$$

$$\cos \theta = \frac{AB \cos(20^\circ) - W}{F} = \frac{AB \cos(20^\circ)}{F} - \frac{W}{F}$$

$$\textcircled{2} \quad \cos \theta = \frac{AB \cos(20^\circ)}{F} - 1 \rightarrow \frac{F(\cos \theta + 1)}{\cos(20^\circ)} = AB$$

$$F = 20 \text{ lb}$$

$$\frac{F \sin \theta}{\sin(20^\circ)} = \frac{F(\cos \theta + 1)}{\cos(20^\circ)}$$

$$\frac{20}{\sin(20^\circ)} \sin \theta = \frac{20}{\cos(20^\circ)} (\cos \theta + 1)$$

$$58.5 \sin \theta = 21.3 (\cos \theta + 1)$$

$$58.5 \sin \theta = 21.3 \cos \theta + 21.3$$

$$5.85 \sin \theta = 2.13 \cos \theta + 2.13 \quad \text{--- Graph}$$

$$\theta = 40.1^\circ$$

4. Force System Resultants

Tuesday, September 3, 2024 9:39 AM

* Shift from a particle analysis to a rigid body analysis (bodies that do not deform under loading)

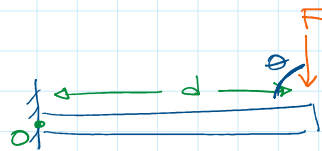
* Leverage

* Application of force matters (position of force on body)

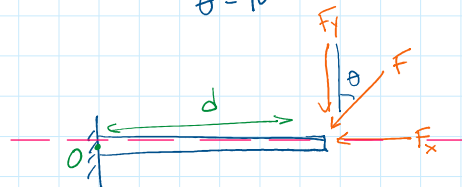
* Bending Moment

$$M_o = F \cdot d \text{ [N}\cdot\text{m]}$$

Perpendicular Vectors
(Force & Position)



$$\theta = 90^\circ$$



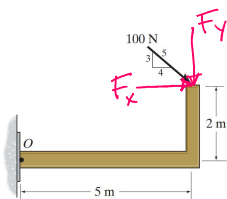
M_o due to F_x is zero because F_x 's line of action goes through point O.

$$M_o \neq F \cdot d$$

$$M_o = F_y \cdot d$$



F4-1. Determine the moment of the force about point O.

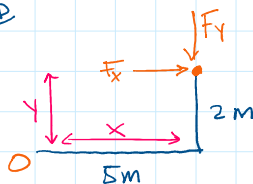


Prob. F4-1

$$M_o = F \cdot d$$

$$F_x = 80 \text{ N} \quad F_y = 60 \text{ N}$$

FBD



* perpendicular distance to F_x is y (from O)

* perpendicular distance to F_y is x (from O)

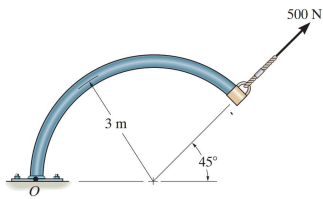
$$M_o = -F_x \cdot y - F_y \cdot x$$

$$M_o = (-80 \text{ N})(2 \text{ m}) - (60 \text{ N})(5 \text{ m})$$

$$= -160 \text{ N}\cdot\text{m} - 300 \text{ N}\cdot\text{m}$$

$$M_o = 460 \text{ N}\cdot\text{m} \text{ cw}$$

F4-6. Determine the moment of the force about point O.



Prob. F4-6

$$\cos(45^\circ) = \frac{a}{3} \quad a = 3 \cos(45^\circ)$$

$$\sin(45^\circ) = \frac{y}{3} \quad y = 3 \sin(45^\circ)$$

$$x = 3 + 3 \cos(45^\circ)$$

$$y = 3 \sin(45^\circ)$$

$$\cos(45^\circ) = \frac{F_x}{F}$$

$$F_x = F \cos(45^\circ)$$

$$F_y = F \sin(45^\circ)$$

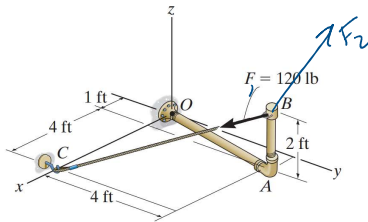
$$M_O = \underbrace{(3 + 3 \cos(45^\circ))}_{x} \underbrace{F_y}_{\text{ccw}} - \underbrace{3 \sin(45^\circ)}_{y} \underbrace{F_x}_{\text{ccw}}$$

$$M_O = (3 + 3 \cos(45^\circ)) F \sin(45^\circ) - 3 \sin(45^\circ) F \cos(45^\circ)$$

$$M_O = (3 + 3 \cos(45^\circ)) 500 \sin(45^\circ) - 3 \sin(45^\circ) (500) \cos(45^\circ)$$

$$M_O = 1060.67 \text{ N}\cdot\text{m} \quad \text{ccw}$$

F4-11. Determine the moment of force $\mathbf{F}$ about point O. Express the result as a Cartesian vector.



Prob. F4-11

$$\vec{M}_O = \vec{r}_{OB} \times \vec{F}$$

* cross product

$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$$

$$\vec{r}_{OB} = 1 \hat{i} + 4 \hat{j} + 2 \hat{k}$$

$$\vec{F} = 120 \hat{u}_{BC}$$

$$\vec{r}_{OC} = 5 \hat{i}$$

$$\vec{r}_{BC} = 5 \hat{i} - [1 \hat{i} + 4 \hat{j} + 2 \hat{k}]$$

$$\vec{r}_{BC} = 4 \hat{i} - 4 \hat{j} - 2 \hat{k} - [1 \hat{i}]$$

$$r_{BC} = \sqrt{4^2 + (-4)^2 + (-2)^2} = \sqrt{36} = 6 \text{ ft}$$

$$\hat{u}_{BC} = \frac{4}{6} \hat{i} - \frac{4}{6} \hat{j} - \frac{2}{6} \hat{k}$$

$$\hat{u}_{BC} = \frac{2}{3} \hat{i} - \frac{2}{3} \hat{j} - \frac{1}{3} \hat{k}$$

$$\vec{F} = F \cdot \hat{u}_{BC} = 120 \left(\frac{2}{3} \hat{i} - \frac{2}{3} \hat{j} - \frac{1}{3} \hat{k} \right) - [16]$$

$$\vec{F} = \underbrace{80 \hat{i}}_{F_x} - \underbrace{80 \hat{j}}_{F_y} - \underbrace{40 \hat{k}}_{F_z} - [16]$$

$$\vec{M}_O = \vec{r} \times \vec{F}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

$$+ (a_y b_z - a_z b_y) \hat{i}$$

$$- (a_x b_z - a_z b_x) \hat{j}$$

$$-(a_x b_z - a_z b_x) \hat{j}$$

$$(a_x b_y - a_y b_x) \hat{k}$$

$$\vec{M}_O = \vec{r}_{OB} \times \vec{F}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 2 \\ 80 & -80 & -40 \end{vmatrix}$$

$$\vec{M}_O = (-160 + 160) \hat{i}$$

$$- [1(-40) - 2(80)] \hat{j}$$

$$+ [1(-80) - 4(80)] \hat{k}$$

$$\vec{M}_O = 200 \hat{j} - 400 \hat{k} \text{ lb}\cdot\text{ft}$$

$$(\vec{M}_O)_1 = \vec{r} \times \vec{F}_1$$

$$(\vec{M}_O)_2 = \vec{r} \times \vec{F}_2$$

$$\vec{F}_R = \vec{F}_1 + \vec{F}_2$$

$$\vec{M}_O = (\vec{M}_O)_1 + (\vec{M}_O)_2$$

$$\vec{M}_O = \vec{r} \times \vec{F}_R$$

F4-12. If the two forces $\mathbf{F}_1 = \{100\mathbf{i} - 120\mathbf{j} + 75\mathbf{k}\}$ lb and $\mathbf{F}_2 = \{-200\mathbf{i} + 250\mathbf{j} + 100\mathbf{k}\}$ lb act at A, determine the resultant moment produced by these forces about point O. Express the result as a Cartesian vector.

$$\vec{F}_1 = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$

$$\vec{F}_1 = 100\hat{i} - 120\hat{j} + 75\hat{k}$$

$$\vec{F}_2 = -200\hat{i} + 250\hat{j} + 100\hat{k}$$

$$\vec{F}_R = -100\hat{i} + 130\hat{j} + 175\hat{k}$$

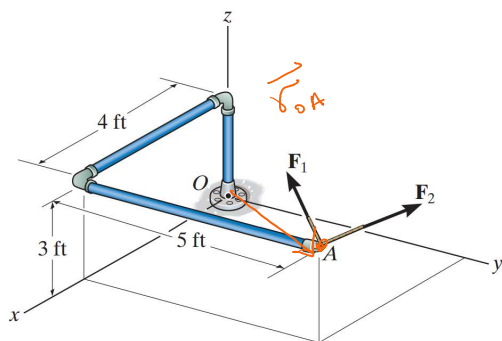
$$\vec{r}_{OA} = 4\hat{i} + 5\hat{j} + 3\hat{k}$$

$$\vec{M}_O = \vec{r}_{OA} \times \vec{F}_R$$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 5 & 3 \\ -100 & 130 & 175 \end{vmatrix}$$

$$[(5)(175) - 3(130)] \hat{i}$$

$$- [4(175) - 3(-100)] \hat{j}$$

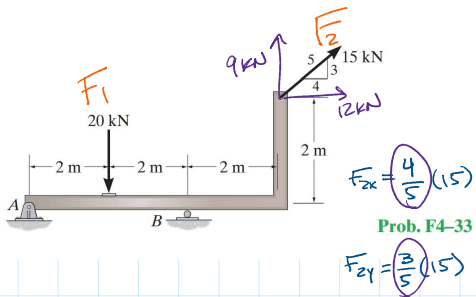


Prob. F4-12

$$[4(130) - 5(-100)] \hat{k}$$

$$\vec{M}_0 = 485\hat{i} - 1000\hat{j} + 1020\hat{k} \text{ [lb}\cdot\text{ft]}$$

F4-33. Replace the loading system by an equivalent resultant force and specify where the resultant's line of action intersects the horizontal segment of the member measured from A.

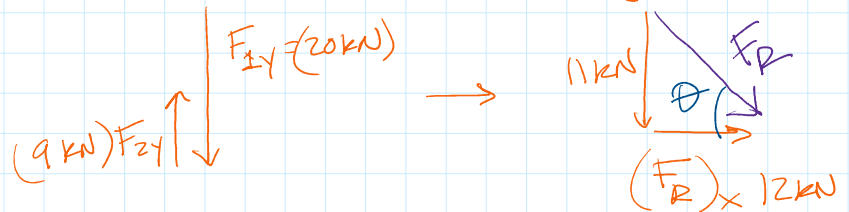


$$F_{1x} = 0$$

$$F_{1y} = 20 \text{ kN}$$

$$F_{2x} = 12 \text{ kN} = \frac{4}{5}(15 \text{ kN})$$

$$F_{2y} = 9 \text{ kN} = \frac{3}{5}(15 \text{ kN})$$



$$F_R^2 = (F_R)_x^2 + (F_R)_y^2$$

$$F_R = \sqrt{11^2 + 12^2} = \sqrt{121 + 144} = \sqrt{265}$$

$$F_R = 16.28 \text{ kN}$$

$$\theta = \tan^{-1}\left(\frac{11}{12}\right) \quad \theta = 42.5^\circ$$

* How far should F_R be from point A?

$\sum M_A = (M_A)_R \rightarrow$ The sum of the moments about A due to F_1 & F_2 should equal the moment about A due to F_R .

$$(M_A)_1 = -x_1 F_{1y} = -(2\text{m})(20\text{kN}) = -40 \text{ kN}\cdot\text{m}$$

$$(M_A)_2 = x_2 F_{2y} - y_2 F_{2x}$$

$$= (6\text{m})(9\text{kN}) - (2\text{m})(12\text{kN}) = 54 \text{ kN}\cdot\text{m} - 24 \text{ kN}\cdot\text{m}$$

$$= 30 \text{ kN}\cdot\text{m}$$

$$+\circlearrowleft \sum M_A = -40 \text{ kN}\cdot\text{m} + 30 \text{ kN}\cdot\text{m} = -10 \text{ kN}\cdot\text{m}$$

$$\sum M_A = 10 \text{ kN}\cdot\text{m} \quad \text{CW}$$

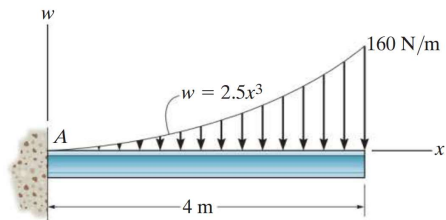
$$\sum M_A = (M_A)_R = x(F_R)_y$$

only account for component that is vertical & perpendicular to x.

$$\frac{10 \text{ kN}\cdot\text{m}}{11 \text{ kN}} = x \frac{(11 \text{ kN})}{11 \text{ kN}}$$

$$x = 0.91 \text{ m}$$

F4-42. Determine the resultant force and specify where it acts on the beam measured from A.



Prob. F4-42

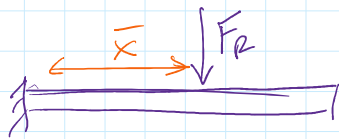
$$F_R = \int_L w(x) dx$$

$$\bar{x} = \frac{\int_L x \cdot w(x) dx}{F_R}$$

equilibrium analysis

$$F_R = \int_0^4 2.5x^3 dx = \frac{5}{8} x^4 \Big|_0^4 = \frac{5}{8} (4)^4 - \frac{5}{8} (0)^4$$

$$F_R = 160 \text{ N}$$



$$\bar{x} = \frac{\int_L x \cdot w dx}{F_R}$$

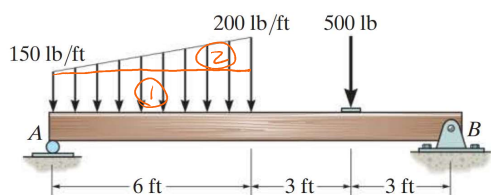
$$\bar{x} = \frac{\int_0^4 x (2.5x^3) dx}{F_R} = \frac{\int_0^4 2.5x^4 dx}{F_R}$$

$$\bar{x} = \frac{\frac{1}{2} x^5 \Big|_0^4}{F_R} = \frac{\frac{1}{2} (4)^5 - 0}{160}$$

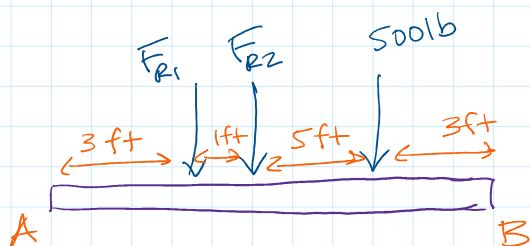
$$\bar{x} = 3.2 \text{ m}$$

F4-40. Determine the resultant force and specify where it acts on the beam measured from A.

& Get the reactions @ the supports.



Prob. F4-40



$$F_{R1} = 150 \frac{\text{lb}}{\text{ft}} (6 \text{ ft})$$

$$F_{R1} = 900 \text{ lb}$$

$$F_{R2} = \frac{1}{2} (6 \text{ ft}) (50 \frac{\text{lb}}{\text{ft}})$$

$$F_{R2} = 150 \text{ lb}$$

$$F_R = F_{R1} + F_{R2} + 500$$

5. Equilibrium of a Rigid Body

Thursday, September 12, 2024 9:37 AM

- * When a body is in a state of Equilibrium its acceleration is zero.
 - * Rigid Body - a body that does not deform (shape is not altered) due to forces being exerted on it.
 - * Sum of Forces equals zero $\sum F = 0$
 - * Sum of Moments equals zero $\sum M = 0$
- New $\rightarrow$

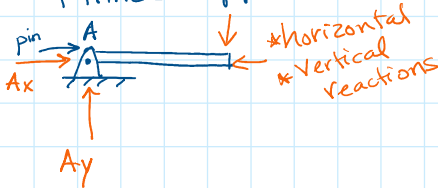
In 2D $\rightarrow \sum F_x = 0 \quad \sum F_y = 0 \quad \sum M_o = 0$
 $\rightarrow \sum F_n = 0 \quad \sum F_t = 0 \quad \sum M_o = 0$



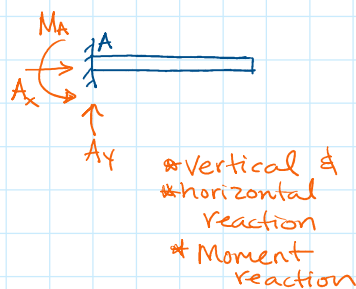
$$F_s = k(l - l_0) \text{ - Spring Force}$$

Types of Supports

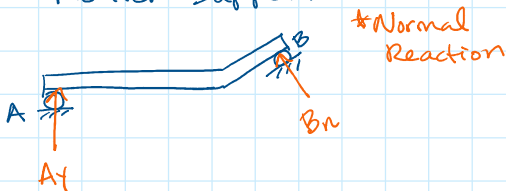
Pinned support



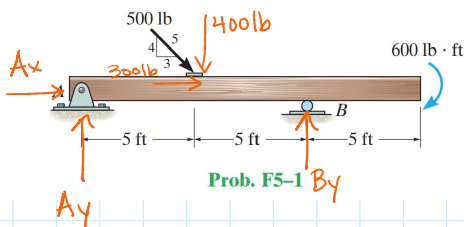
Fixed Support



Roller Support



F5-1. Determine the horizontal and vertical components of reaction at the supports. Neglect the thickness of the beam.



A - pinned
B - roller

$$\begin{aligned} \sum \vec{F} &= 0 \rightarrow \sum F_x = 0 \\ &\quad \sum F_y = 0 \\ \sum M_A &= 0 \end{aligned}$$

$$\sum M_A = -600 \text{ lb}\cdot\text{ft} - (5 \text{ ft})(400 \text{ lb}) + (10 \text{ ft})B_y = 0$$

$$10 B_y = 2600 \quad B_y = 260 \text{ lb} \uparrow$$

$$\sum F_x = A_x + 300 = 0 \quad A_x = -300 \text{ lb}$$

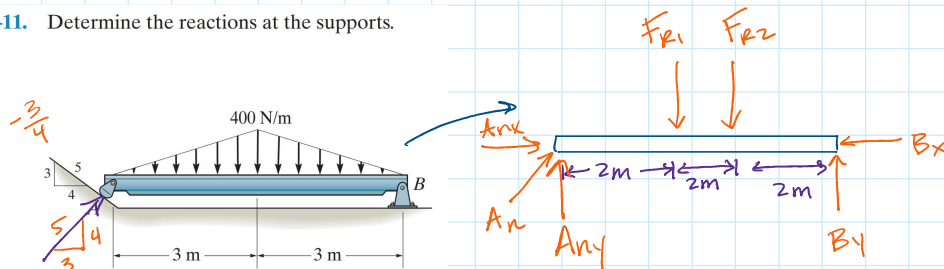
$$A_x = 300 \text{ lb} \leftarrow$$

$$\sum F_y = A_y + B_y - 400 = 0 \rightarrow A_y + 260 - 400 = 0$$

$$A_y = 140 \text{ lb}$$

$$A_y = 140 \text{ lb} \uparrow$$

5-11. Determine the reactions at the supports.



Prob. 5-11

$$F_{R1} = \frac{1}{2} b \cdot h$$

$$F_{R1} = \frac{1}{2} (3 \text{ m})(400 \text{ N/m})$$

$$F_{R1} = 600 \text{ N} = F_{R2}$$

⊥ lines have opposite reciprocal slopes

$$A_{nx} = \frac{3}{5} A_n$$

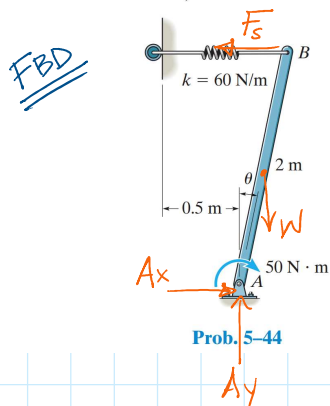
$$A_{ny} = \frac{4}{5} A_n$$

$$\sum M_B = 2(F_{R2}) + 4(F_{R1}) - 6\left(\frac{24}{5} A_n\right) = 0$$

$$\frac{24}{5} A_n = 2(600) + 4(600)$$

$$A_n = \frac{5}{24} (3600) \quad A_n = 750 \text{ N normal to surface}$$

*5-44. The 10-kg uniform rod is pinned at end A. If it is also subjected to a couple moment of 50 N·m, determine the smallest angle θ for equilibrium. The spring is unstretched when $\theta = 0$, and has a stiffness of $k = 60 \text{ N/m}$.



Prob. 5-44

Knowns

$$M = -50 \text{ N}\cdot\text{m}$$

$$k = 60 \text{ N/m}$$

$$m = 10 \text{ kg}$$

$$l_r = 2 \text{ m}$$

Pinned support @ A

$$l_0 = 0.5 \text{ m}$$

* Spring remains horizontal

$$\sum F_x = 0 \quad \sum F_y = 0 \quad \sum M_A = 0$$

$$F_s = k(l - l_0)$$

Unknowns

unknowns

A_x, A_y, F_s, θ, l , can relate l & θ using geometry

$$\sum M_A = 0$$

$$\sum M_A = -\frac{l_r \sin \theta}{2} \cdot w - 50 + l \cos \theta F_s = 0$$

$$l_r \cos \theta \left[K \underbrace{(l - l_0)}_{l_r \sin \theta} \right] - \frac{l_r \sin \theta}{2} \cdot \underbrace{w}_{mg} = 50$$

$$l_r^2 \cdot K \cos \theta \sin \theta - \frac{l_r}{2} \sin \theta \cdot m \cdot g = 50$$

divided
by 10

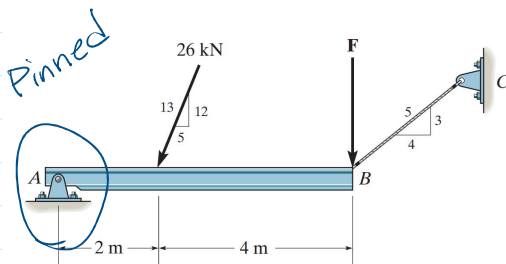
$$(2)^2 (\cancel{60}) \cos \theta \sin \theta - \frac{(2)}{2} (\cancel{10}) (9.81) \sin \theta = \cancel{50}$$

$$24 \cos \theta \sin \theta = 9.81 \sin \theta + 5$$

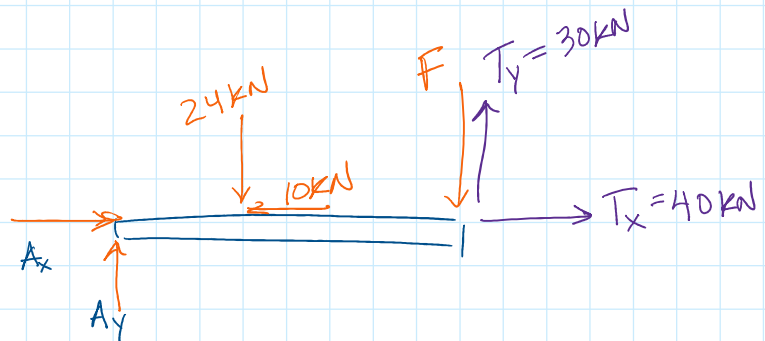
$$\theta = 0.429 \text{ rad} \left(\frac{180^\circ}{\pi \text{ rad}} \right)$$

$$\theta = 24.57^\circ$$

*5-48. If rope BC will fail when the tension becomes 50 kN, determine the greatest vertical load F that can be applied to the beam at B. What is the magnitude of the reaction at A for this loading? Neglect the thickness of the beam.



Probs. 5-47/48



Too many unknowns

$$\sum F_x = 0 \quad \sum F_y = 0 \quad \sum M_A = 0$$

$$\sum M_A = 0$$

$$-(2)(24) - (6)(F) + (6)(30 \text{ kN}) = 0$$

$$A_x = 30 \text{ kN} \leftarrow$$

$$A_y = 16 \text{ kN} \uparrow$$

$$-(2)(24) - (6)(F) + (6)(30 \text{ kN}) = 0$$

$$6F = 6(30) - 2(24)$$

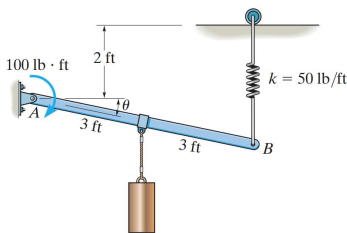
$$6F = 180 - 48$$

$$F = 30 - 8$$

$$F = 22 \text{ kN}$$

X 0.90

5-59. The rod supports a weight of 200 lb and is pinned at its end *A*. If it is also subjected to a couple moment of 100 lb · ft, determine the angle θ for equilibrium. The spring has an unstretched length of 2 ft and a stiffness of $k = 50 \text{ lb/ft}$.

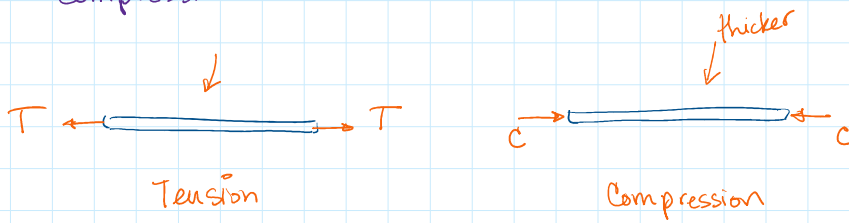


Prob. 5-59

6. Structural Analysis

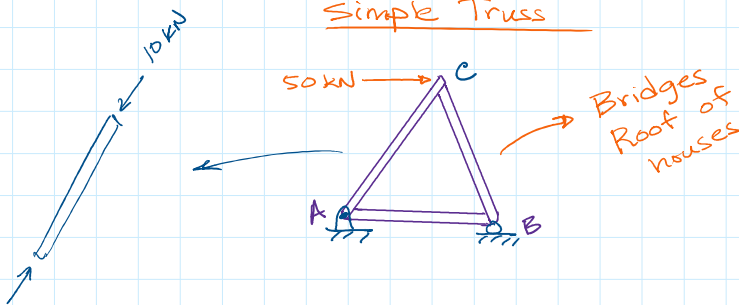
Thursday, September 26, 2024 9:42 AM

- * Analyzing trusses (consists of slender members)
- * Slender members can be in a state of tension or compression.



Stress-strain analysis

Simple Truss



Method of Joints

- * Joints are in equilibrium
- * All forces on joint must cancel each other out

$$\sum F_x = 0 \quad \sum F_y = 0$$

Method of Sections

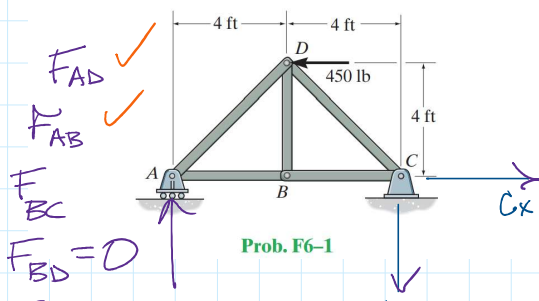
- * Draw a line that goes through a few members and consider a section of truss.

- * Section is in equilibrium

$$\sum F_x = 0 \quad \sum F_y = 0$$

$$\sum M = 0$$

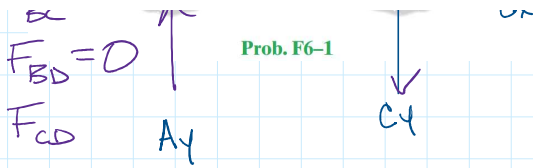
F6-1. Determine the force in each member of the truss. State if the members are in tension or compression.



Prob. F6-1

- * Consider the truss to be uniform rigid body
- * $\sum F_x = 0$, $\sum F_y = 0$
 $\sum M_c = 0$

- * Start with $\sum M_c$



Prob. F6-1

* Start with ΣM_C because you can neglect two unknowns (C_x & C_y). A_y is only unknown.

$$\uparrow \Sigma M_C = -8A_y + 4(450) = 0$$

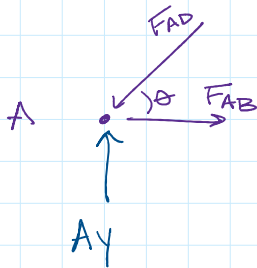
$$A_y = 225 \text{ lb } \uparrow$$

$$C_y = 225 \text{ lb } \downarrow$$

$$C_x = 450 \text{ lb } \rightarrow$$

Method of Joints

Joint A



$$\theta = 45^\circ$$

$$\Sigma F_x = 0 \quad \Sigma F_y = 0$$

$$\Sigma F_y = A_y - F_{AD} \sin \theta = 0$$

$$F_{AD} = \frac{A_y}{\sin \theta}$$

$$F_{AD} = \frac{225}{\sin(45^\circ)} = 318.2 \text{ lb}$$

$$F_{AD} = 318.2 \text{ lb (C)}$$

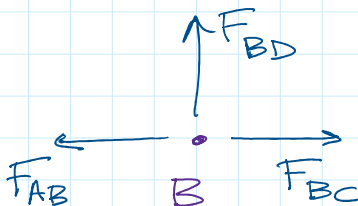
$$\Sigma F_x = F_{AB} - F_{AD} \cos \theta = 0$$

$$F_{AB} = F_{AD} \cos \theta$$

$$F_{AB} = (318.2) \cos(45^\circ) = 225 \text{ lb}$$

$$F_{AB} = 225 \text{ lb (T)}$$

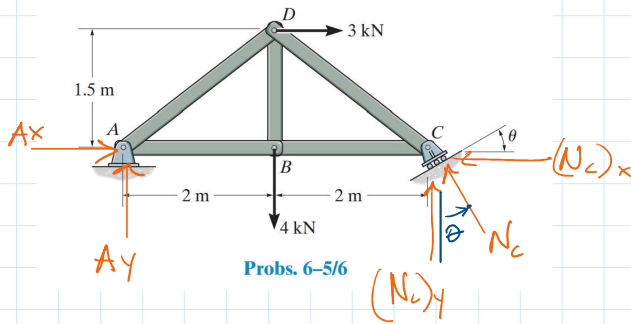
Joint B



$$\Sigma F_x = 0 \quad F_{BC} - F_{AB} = 0$$

$$F_{BC} = \dots$$

6-6. Determine the force in each member of the truss, and state if the members are in tension or compression. Set $\theta = 30^\circ$.



$$\sum F_x = 0$$

$$A_x + 3 - (N_c)_x = 0$$

$$A_x = (N_c)_x - 3$$

$$A_x = N_c \sin \theta - 3$$

$$A_x = 3.6 \sin(30^\circ) - 3$$

$$A_x = 1.8 - 3$$

$$A_x = 1.2 \text{ kN} \leftarrow$$

$$(1.5 \text{ m})(3 \text{ kN}) + (2 \text{ m})(4 \text{ kN}) = (4 \text{ m}) N_c \cos \theta$$

$$4.5 + 8 = 4 N_c \cos(30^\circ)$$

$$\frac{12.5}{4} = N_c \cos(30^\circ)$$

$$N_c = \frac{3.125}{\cos(30^\circ)}$$

$$N_c = 3.6 \text{ kN}$$

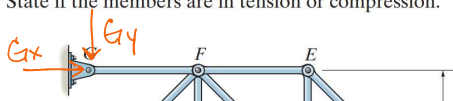
$$\sum F_y = 0 \Rightarrow A_y - 4 + (N_c)_y = 0$$

$$A_y = 4 - N_c \cos \theta$$

$$A_y = 4 - 3.6 \cos(30^\circ)$$

$$A_y = 0.875 \text{ kN} \uparrow$$

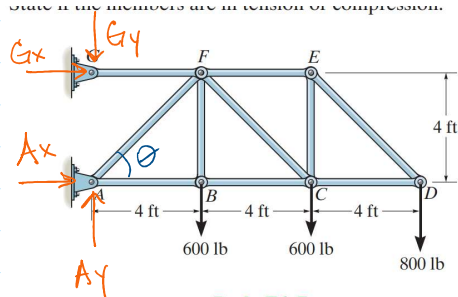
F6-7. Determine the force in members BC, CF, and FE. State if the members are in tension or compression.



$$\theta = 45^\circ$$

* First solve for reaction @ supports.

State if the members are in tension or compression.



Prob. F6-7

* First solve for reaction @ supports.

$$\uparrow \sum M_A = 0 \quad \text{neglect } A_x, A_y, G_y$$

$$\begin{aligned} \uparrow \sum M_A &= -4(600) - 8(600) - 12(800) - 4G_x = 0 \\ -2400 - 4800 - 9600 - 4G_x &= 0 \\ -16,800 &= 4G_x \end{aligned}$$

$$G_x = -4200 \text{ lb}$$

$$G_x = 4200 \text{ lb} \leftarrow$$

$$\rightarrow \sum F_x = 0$$

$$A_x = 4200 \text{ lb} \rightarrow$$

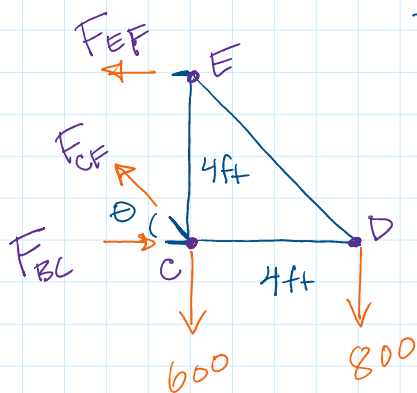
$$\uparrow \sum F_y = 0 \rightarrow A_y - G_y - 600 - 600 - 800 = 0$$

$$A_y - G_y = 2000 \text{ lb}$$

$$\uparrow \sum M_C = 0 \quad -4(800) + 4 F_{EF} = 0$$

$$F_{EF} = 800 \text{ lb}$$

$$F_{EF} = 800 \text{ lb (T)}$$



$$\theta = 45^\circ$$

$$\uparrow \sum F_y = 0$$

$$\begin{aligned} F_{CF} \sin(45^\circ) - 600 - 800 &= 0 \\ F_{CF} \sin(45^\circ) &= 1400 \end{aligned}$$

$$F_{CF} = \frac{1400}{\sin(45^\circ)} = 1980 \text{ lb (T)}$$

$$\rightarrow \sum F_x = 0 \quad -F_{EF} - F_{CF} \cos(45^\circ) + F_{BC} = 0$$

$$\begin{aligned} +800 + 1980 \cos(45^\circ) &= F_{BC} \\ +800 + 1400 &= F_{BC} \end{aligned}$$

$$F_{BC} = +2200$$

$$F_{BC} = 2200 \text{ lb (C)}$$

6-38. Determine the force in members BC, BE, and EF of the truss and state if these members are in tension or compression. Set $P_1 = 6 \text{ kN}$, $P_2 = 9 \text{ kN}$, and $P_3 = 12 \text{ kN}$.

$$\sum M_A = 0$$

or compression. Set $P_1 = 6 \text{ kN}$, $P_2 = 9 \text{ kN}$, and $P_3 = 12 \text{ kN}$.

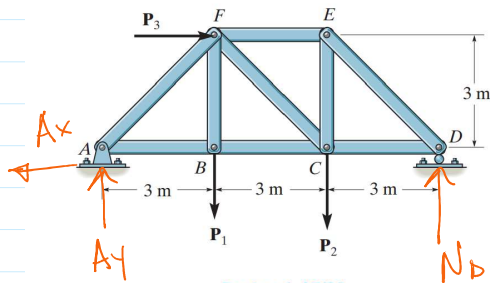
$$\sum M_A = 0$$

$$9 N_D = 3P_1 + 6P_2 + 3P_3$$



$$N_D = \frac{3(6) + 6(9) + 3(12)}{9}$$

$$N_D = 12 \text{ kN } \uparrow$$



Probs. 6-37/38

$$A_x = 12 \text{ kN } \leftarrow$$

$$A_y = 3 \text{ kN } \uparrow$$

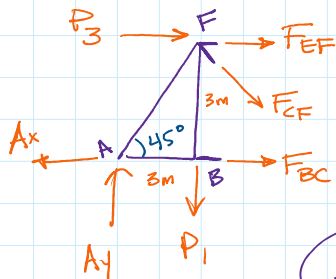
$$+\circlearrowleft \sum M_F = 0$$

Don't account P_1, P_3, F_{EF}, F_{CF}

$$3F_{BC} - 3A_x - 3A_y = 0$$

$$F_{BC} = \frac{3(12+3)}{3} = 15 \text{ kN}$$

$$F_{BC} = 15 \text{ kN (T)}$$



$$+\rightarrow \sum F_x = 0$$

$$F_{BC} + F_{CF} \cos(45^\circ) + F_{EF} + P_3 - A_x = 0$$

$$F_{EF} + F_{CF} \cos(45^\circ) = A_x - P_3 - F_{BC}$$

$$F_{EF} + F_{CF} \cos(45^\circ) = 12 - 12 - 15$$

$$F_{EF} + F_{CF} \cos(45^\circ) = -15$$

$$F_{EF} + (-4.24) \cos(45^\circ) = -15$$

$$F_{EF} - 3 = -15$$

$$F_{EF} = -12 \text{ kN}$$

$$F_{EF} = 12 \text{ kN (C)}$$

$$+\uparrow \sum F_y = 0$$

$$A_y - P_1 - F_{CF} \sin(45^\circ) = 0$$

$$F_{CF} \sin(45^\circ) = A_y - P_1$$

$$F_{CF} \sin(45^\circ) = 3 - 6$$

$$F_{CF} = \frac{-3}{\sin(45^\circ)} = -4.24 \text{ kN}$$

$$F_{CF} = 4.24 \text{ kN (C)}$$

8. Friction

Thursday, October 10, 2024 9:42 AM

- * Equilibrium of a Rigid Body
- * Accounting for frictional forces
- * Dry Friction (Reaction force)
- * Typically between two surfaces

* Impending Motion

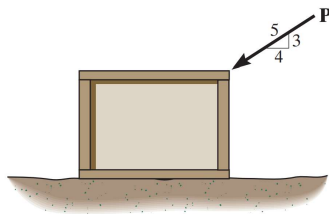
* Object is on the verge of slipping

$$F_s = \mu_s N$$

F_s → Static
 μ_s → coefficient of static friction (unitless)
 N → Normal force (lb or N)

μ_s - coefficient of static friction $\mu_s < 1$
 ↳ obtained experimentally

F8-1. Determine the friction developed between the 50-kg crate and the ground if a) $P = 200$ N, and b) $P = 400$ N. The coefficients of static and kinetic friction between the crate and the ground are $\mu_s = 0.3$ and $\mu_k = 0.2$.



Prob. F8-1

$$F_s = \mu_s N$$

$$F_s = \mu_s (W + \frac{3}{5}P)$$

$$F_s = \mu_s (mg + \frac{3}{5}P)$$

$$F_s = 0.3 [(50)(9.81) + \frac{3}{5}(200)]$$

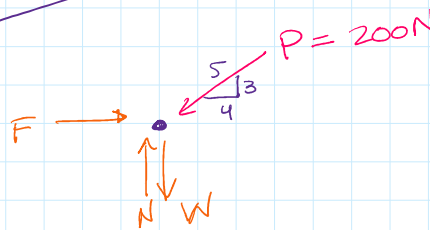
$$F_s = 0.3 [490.5 + 120]$$

$$F_s = 0.3 [610.5]$$

$$F_s = 183.15 \text{ N}$$

Friction right before slips.

Point Particle



$$N = W + P_y$$

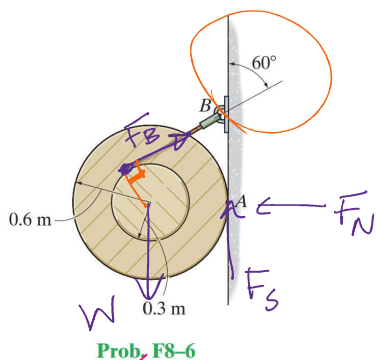
$$N = W + \frac{3}{5}P$$

$$F = \frac{4}{5}P = \frac{4}{5}(200 \text{ N})$$

$$F = 160 \text{ N} \text{ - Maybe}$$

↳ Friction needed for equilibrium

F8-6. Determine the minimum coefficient of static friction between the uniform 50-kg spool and the wall so that the spool does not slip.



Prob. F8-6

✓
 $F_s = \mu_s F_N$

$\sum M_O$
 $0.3 F_B = 0.6 F_s$
 $F_B = 2 F_s$ (1)

$\sum F_y$
 $F_s - W + F_B \cos(60^\circ) = 0$ (2)

$\sum F_x$
 $F_B \sin(60^\circ) - F_N = 0$ (3)

(2a) $F_s - W + 2 F_s \cos(60^\circ) = 0$

(3a) $2 F_s \sin(60^\circ) = F_N$

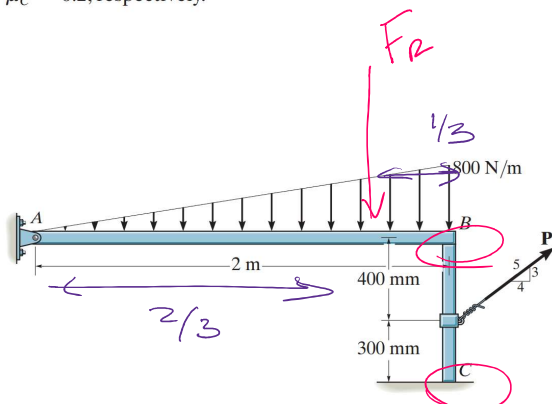
(2b) $F_s (1 + 2 \cos(60^\circ)) = W$

$\mu_s F_N = \frac{W}{1 + 2 \cos(60^\circ)}$

(3b) $2 \mu_s F_N \sin(60^\circ) = F_N$

$\mu_s = \frac{1}{2 \sin(60^\circ)} = \frac{1}{2(\frac{\sqrt{3}}{2})} = \frac{\sqrt{3}}{3} = 0.577$

8-45. The beam AB has a negligible mass and thickness and is subjected to a triangular distributed loading. It is supported at one end by a pin and at the other end by a post having a mass of 50 kg and negligible thickness. Determine the minimum force P needed to move the post. The coefficients of static friction at B and C are $\mu_B = 0.4$ and $\mu_C = 0.2$, respectively.



Prob. 8-45

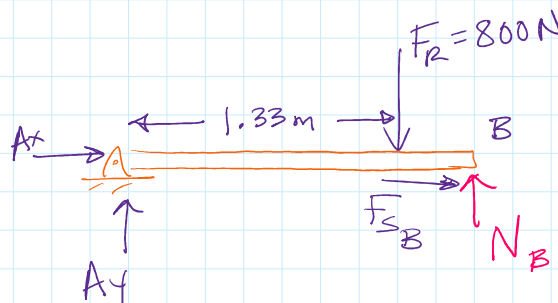
$F_s = \mu_s N$

$F_R = \frac{1}{2} b h$

$F_R = \frac{1}{2} (2 \text{ m}) (800 \text{ N/m})$

$F_R = 800 \text{ N}$

$\bar{x} = 2 \text{ m} (\frac{1}{3}) = \frac{2}{3} \text{ m} = 0.667 \text{ m}$
 to the left of B



$$F_s = \mu_s N$$

$\uparrow$
 A_y

$\uparrow$
 N_B

$$\sum M_A = 0$$

$$-\frac{4}{3}(800) + 2N_B = 0$$

$$N_B = \frac{\frac{4}{3}(800)}{2}$$

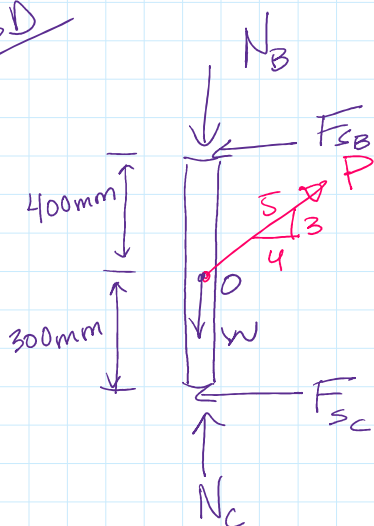
$$N_B = \frac{1600}{3} \text{ N}$$

$$N_B = 533\frac{1}{3} \text{ N} \uparrow$$

$$F_{sB} = \mu_B N_B$$

only when there's impending motions

FBD



$$\sum F_y = 0$$

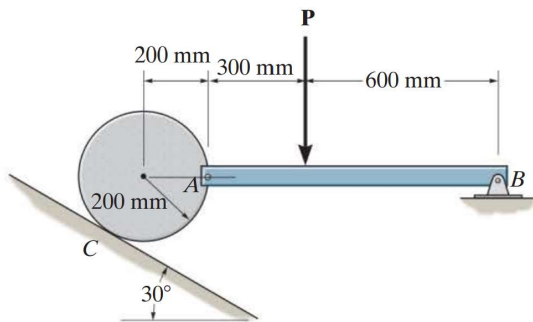
$$N_C + \frac{3}{5}P - W - N_B = 0$$

$$N_C = W + N_B - \frac{3}{5}P$$

$$\sum F_x = 0$$

$$\sum M_O = 0$$

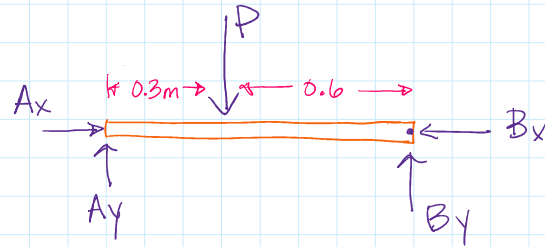
R8-3. A 35-kg disk rests on an inclined surface for which $\mu_s = 0.2$. Determine the maximum vertical force $\mathbf{P}$ that may be applied to bar AB without causing the disk to slip at C . Neglect the mass of the bar.



Prob. R8-3

$$M_d = 35 \text{ kg}$$

$$\mu_s = 0.2$$



$$\sum M_B = 0$$

$$0.9 A_y = 0.6 P$$

$$A_y = \frac{2}{3} P$$

$$P = \frac{3}{2} A_y$$

$$\sum F_x = 0$$

$$A_x = B_x$$

$$\sum F_y = 0$$

$$A_y + B_y = P$$

FBD disk $r = 0.2 \text{ m}$

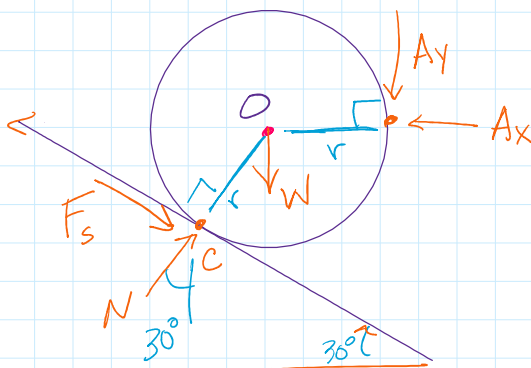
$$\sum M_O = 0$$

$$-r A_y + r F_s = 0$$

$$A_y = F_s$$

$$A_y = \mu_s N$$

$$P = \frac{3}{2} \mu_s N$$

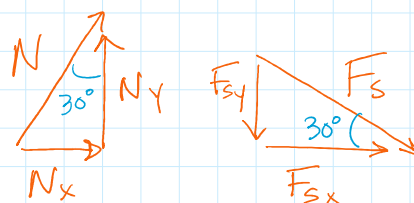


$$N_x = N \sin(30^\circ)$$

$$N_y = N \cos(30^\circ)$$

$$F_{sx} = F_s \cos(30^\circ)$$

$$F_{sy} = F_s \sin(30^\circ)$$



$$\sum F_x = 0$$

$$F_{sy} = F_s \sin(30^\circ)$$

$$\Sigma F_x = 0$$

$$N_x + F_{sx} - A_x = 0$$

$$\Sigma F_y = 0$$

$$N_y - F_{sy} - W - A_y = 0$$

$$N \cos(30^\circ) - \underline{F_s} \sin(30^\circ) - mg - A_y = 0$$

$$N \cos(30^\circ) - \mu_s N \sin(30^\circ) - mg - A_y \downarrow = 0$$

$$N \cos(30^\circ) - \mu_s N \sin(30^\circ) - \mu_s N = mg$$

$$N [\cos(30^\circ) - \mu_s \sin(30^\circ) - \mu_s] = mg$$

$$N = \frac{mg}{\cos(30^\circ) - \mu_s \sin(30^\circ) - \mu_s}$$